

PHYS 708 – Assignment 2 (Due date: 6 October 2025)

Problem

(a) Obtain the spectrum of particles accelerated by diffusive shock acceleration by solving the following transport equation in the rest frame of the shock,

$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial x} \left(p \frac{\partial f}{\partial p} \right) + Q(x, p) = 0$$

where $D(p)$ is the diffusion coefficient assumed to be same on both sides of the shock front, x denotes the position with respect to the shock front (positive towards the downstream region and negative towards upstream), $f(x, p)$ is the phase space distribution function for particles with momentum p , u is the velocity of the gas (different for the different regions), and $Q(x, p)$ is the phase space density of thermal seed particles injected at the shock.

(b) Using the solution obtained, show that the particle spectrum follows a power-law $N(p) \propto p^{-\Gamma}$ with an index $\Gamma = 2$ for strong shocks.

Hint: Follow the steps described on the lecture slides. The solutions in the upstream and the downstream regions should be in the forms,

$$f(p) = f_0, \quad \text{downstream (x > 0)}$$

$$f(p) = f_0 e^{\frac{u_1}{D(p)} x}, \quad \text{upstream (x < 0)}$$

with

$$f_0 = \frac{\gamma n_i}{4\pi p_i^3} \left(\frac{p}{p_i} \right)^{-\gamma}$$

where

$$\gamma = \frac{3u_1}{u_1 - u_2} = \frac{3r}{r - 1}$$

u_1 and u_2 represents the gas velocities in the upstream and downstream regions respectively with respect to the shock front. n_i and p_i are the number density and the momentum of the thermal seed particles injected at the shock. $r = u_1/u_2$ is the shock compression ratio. Refer to the lecture slides for more details.